

Supplementary Material for

Robust Non-Invasive Monitoring of Cumulative Tibia Load in Male Recreational Runners Using Multi-Modal Wearables and Deep Learning

S I. Supplementary Tables

TABLE S1. Definition of clinical data scenarios and sampling distribution configuration for joint robustness training.

Fault level	Scenario ID	Scenario description	No. of sEMG channels	IMU status	Sampling weight	Relative proportion
I. Ideal state	S1	Ideal acquisition state	8 (Full)	ON	25	17.50%
II. Minor fault	S2	Single electrode detachment	7	ON	5	3.50%
	S3	Double electrode detachment	6	ON	5	3.50%
	S4	Multi-electrode interference	5	ON	5	3.50%
	S5	Half electrode failure	4	ON	5	3.50%
III. Severe fault	S6	Loss of core muscle groups	3	ON	9	6.30%
	S7	Minimal sEMG features	2	ON	9	6.30%
	S8	Single muscle limit inference	1	ON	9	6.30%
	S9	Complete sEMG missing	0	ON	9	6.30%
IV. Modality loss	S10	IMU missing	8	OFF	5	3.50%
V. Dual faults	S11	IMU failure + single electrode detachment	7	OFF	9	6.30%
	S12	IMU failure + double electrode detachment	6	OFF	9	6.30%
	S13	IMU failure + multi-electrode interference	5	OFF	9	6.30%
	S14	IMU failure + half electrode failure	4	OFF	9	6.30%
VI. Extreme challenge	S15	Catastrophic signal loss	3	OFF	7	4.90%
	S16	Near system collapse	2	OFF	7	4.90%
	S17	Worst-case scenario	1	OFF	7	4.90%

Note: sEMG, surface electromyography; IMU, inertial measurement unit.

TABLE S2. Optimal hyperparameter settings for the proposed CNNT and baseline models.

Categories	Hyperparameter	Search Space	CNNT	CNN-BiGRU-Attention	CNN-BiLSTM-HAM	CNN-LSTM	LSTM-MLP	TCN
Training Configuration	Batch Size	[64, 512]	133	118	132	101	104	152
	Optimizer	{Adam, SGD, RMSprop, Adagrad}	Adam	Adam	Adam	RMSprop	Adam	Adam
	Learning Rate	$[1.00 \times 10^{-5}, 1.00 \times 10^{-2}]$ (Log-uniform)	6.02×10^{-5}	1.46×10^{-4}	5.04×10^{-4}	1.48×10^{-4}	2.34×10^{-4}	1.24×10^{-3}
	Weight Decay	$[1.00 \times 10^{-6}, 1.00 \times 10^{-3}]$ (Log-uniform)	3.19×10^{-4}	5.51×10^{-4}	1.05×10^{-4}	6.00×10^{-4}	6.06×10^{-5}	4.21×10^{-4}
Network Architecture	Hidden Dimension (d_{model})	[64, 512]	102	158	122	118	146	128
	Activation Function	{ReLU, Tanh, Sigmoid, GELU}	GELU	ReLU	ReLU	GELU	ReLU	Tanh
	Dropout Rate	[0.1, 0.6]	0.42	0.32	0.34	0.38	0.49	0.51
Feature Extractor (CNN)	CNN Filters	[64, 512] (Hybrid) / [16, 128] (TCN)	126	144	136	142	—	45
	Kernel Size	{3, 5, 10, 15}	3	3	3	3	—	3
	CNN Layers	[1, 3]	2	2	2	1	—	—
Sequence Modeling	Layer Type	—	Transformer	BiGRU	BiLSTM	LSTM	LSTM	TCN
	Number of Layers	[1, 6] (Transformer) / [1, 4] (LSTM/BiGRU/BiLSTM) / [2, 5] (TCN)	2	2	2	2	2	2
	Attention Heads	{2, 4, 8}	2	2	2 (10 channels)	—	—	—
Regressor (MLP)	MLP Hidden Units	[50, 300]	64	102	128	114	108	131
	MLP Layers	[1, 3]	2	2	2	2	2	2

Note: Values indicate the selected hyperparameters for each model after hyperparameter optimization. CNNT, convolutional neural network–transformer; CNN-BiGRU-Attention, convolutional neural network–bidirectional gated recurrent unit with attention; CNN-BiLSTM-HAM, convolutional neural network–bidirectional long short-term memory network with a hybrid attention mechanism; CNN-LSTM, convolutional neural network–long short-term memory network; LSTM-MLP, long short-term memory network with multilayer perceptron; TCN, temporal convolutional network; d_{model} , hidden feature dimension. “Attention Heads” denotes the number of attention heads used in the transformer or attention-based recurrent modules; “2 (10 channels)” indicates a two-head hybrid attention module operating on a 10-channel feature representation. “—” indicates that the corresponding hyperparameter was not applicable to that model.

TABLE S3. Ablation study results quantifying the impact of the SCAM module.

Target	Metric	CNNT	CNNT-SCAM
$\sigma_1^{\text{peak}}(t)$	R^2	0.90 (0.07)	0.90 (0.08)
	RMSE (MPa)	3.42 (1.19)	3.42 (1.30)
	MAE (MPa)	2.63 (0.88)	2.59 (0.88)
	nRMSE	0.08 (0.02)	0.08 (0.02)
σ_1^{GP}	RMSE (MPa)	4.19 (1.71)	4.08 (2.11)
	MAE (MPa)	3.35 (1.30)	3.24 (1.48)
	MAPE (%)	10.92 (3.52)	10.47 (3.60)
	ATE (%)	4.38 (2.01)	4.34 (2.15)
	Mean_ED (mm)	9.47 (2.94)	9.27 (3.00)
	Acc_total	0.81 (0.19)	0.81 (0.19)
	Acc_sector	0.92 (0.11)	0.91 (0.13)
$\sigma_3^{\text{peak}}(t)$	R^2	0.87 (0.13)	0.88 (0.12)
	RMSE (MPa)	6.54 (3.05)	6.39 (2.93)
	MAE (MPa)	4.91 (1.46)	4.79 (1.42)
	nRMSE	0.08 (0.02)	0.08 (0.02)
σ_3^{GP}	RMSE (MPa)	9.27 (7.27)	8.96 (6.63)
	MAE (MPa)	6.96 (3.84)	6.91 (3.28)
	MAPE (%)	10.58 (3.85)	10.67 (3.36)
	ATE (%)	3.89 (1.88)	3.86 (1.86)
	Mean_ED (mm)	8.24 (2.55)	8.27 (2.19)
	Acc_total	0.68 (0.17)	0.69 (0.17)
	Acc_sector	0.79 (0.16)	0.83 (0.16)

Note: Bold font denotes the numerically superior performance. CNNT, convolutional neural network-transformer; CNNT-SCAM, convolutional neural network-transformer with self-correction with adaptive mask; $\sigma_1^{\text{peak}}(t)$, temporal evolution of peak maximum principal stress; $\sigma_3^{\text{peak}}(t)$, temporal evolution of peak minimum principal stress; σ_1^{GP} , global peak maximum principal stress; σ_3^{GP} , global peak minimum principal stress; RMSE, root mean square error; MAE, mean absolute error; nRMSE, normalized root mean square error; MAPE, mean absolute percentage error; ATE, absolute time error; Mean_ED, mean Euclidean distance; Acc_total, 8-zone accuracy; Acc_sector, 4-sector accuracy.

TABLE S4. Performance comparison of models on clean data before and after joint robustness training.

Target	Metric	CNNT-SCAM	CNNT-SCAM-R
$\sigma_1^{\text{peak}}(t)$	R^2	0.90 (0.08)	0.88 (0.06)*
	RMSE (MPa)	3.42 (1.30)	3.71 (1.06)
	MAE (MPa)	2.59 (0.88)	2.90 (0.83)*
	nRMSE	0.08 (0.02)	0.08 (0.02)
σ_1^{GP}	RMSE (MPa)	4.08 (2.11)	4.39 (1.72)
	MAE (MPa)	3.24 (1.48)	3.63 (1.44)
	MAPE (%)	10.47 (3.60)	12.19 (4.48)
	ATE (%)	4.34 (2.15)	4.95 (2.94)
	Mean_ED (mm)	9.27 (3.00)	10.74 (2.73)*
	Acc_total	0.81 (0.19)	0.74 (0.18)
	Acc_sector	0.91 (0.13)	0.87 (0.15)
$\sigma_3^{\text{peak}}(t)$	R^2	0.88 (0.12)	0.87 (0.09)*
	RMSE (MPa)	6.39 (2.93)	6.69 (1.92)*
	MAE (MPa)	4.79 (1.42)	5.28 (1.46)*
	nRMSE	0.08 (0.02)	0.08 (0.02)*
σ_3^{GP}	RMSE (MPa)	8.96 (6.63)	8.85 (3.74)
	MAE (MPa)	6.91 (3.28)	7.17 (2.83)
	MAPE (%)	10.67 (3.36)	11.70 (4.68)
	ATE (%)	3.86 (1.86)	4.54 (2.22)
	Mean_ED (mm)	8.27 (2.19)	9.27 (2.44)**
	Acc_total	0.69 (0.17)	0.67 (0.16)
	Acc_sector	0.83 (0.16)	0.80 (0.14)

Note: Bold font denotes the numerically superior performance. * denotes ($P < 0.05$); ** denotes ($P < 0.005$). CNNT-SCAM, convolutional neural network–transformer with self-correction with adaptive mask; CNNT-SCAM-R, CNNT-SCAM trained via joint robustness training; $\sigma_1^{\text{peak}}(t)$, temporal evolution of peak maximum principal stress; $\sigma_3^{\text{peak}}(t)$, temporal evolution of peak minimum principal stress; σ_1^{GP} , global peak maximum principal stress; σ_3^{GP} , global peak minimum principal stress; RMSE, root mean square error; MAE, mean absolute error; nRMSE, normalized root mean square error; MAPE, mean absolute percentage error; ATE, absolute time error; Mean_ED, mean Euclidean distance; Acc_total, 8-zone accuracy; Acc_sector, 4-sector accuracy.

S II. Supplementary Figures

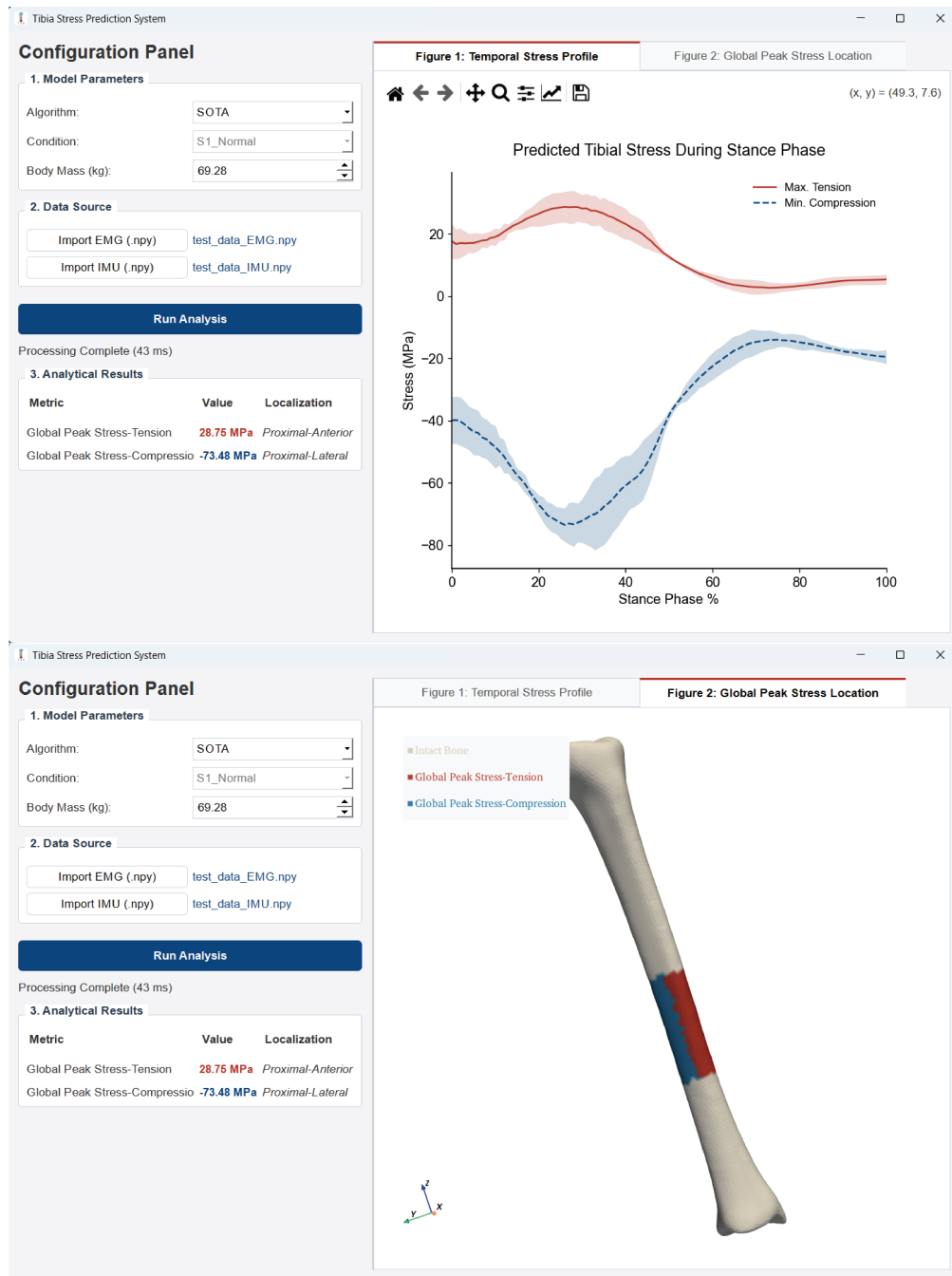


Fig. S1. Graphical user interface of the tibial stress prediction system for clinical deployment. The upper panel displays the temporal stress analysis dashboard. The lower panel presents the 3D anatomical localization dashboard. This view maps the predicted global peak stress locations onto a standardized 3D tibial mesh, visually highlighting the global peak maximum principal stress (red) and global peak minimum principal stress (blue) regions.